Zaiqian "Ting" Chen

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Education

2024–Present **PhD, Statistics**, *Columbia University*, New York, NY.

2022–2024 **MS, Computer Science**, *Columbia University*, New York, NY, GPA: *3.99/4.00*.

2017–2021 **BS, Computer Science (honors), Mathematical Economics, *Summa Cum Laude***, *University of Richmond*, Richmond, VA, GPA: *3.94/4.00*.

Fall 2019 **Visiting Student**, *University of Edinburgh*, Edinburgh, Scotland, GPA: *4.0/4.0* (conv.).

Publications & Preprints

arXiv **Z. Chen**, et al., *The Role of Quantization in Loss Spikes*. arXiv preprint, 2026.

In submission **Z. Chen**, J. Wenger, J. Cunningham, *Scalable Latent Space Bayesian Optimization Through Locality and Parsimony*.

COLM 2026 A. Chun, A. Rao, A. Acherjee, **Z. Chen**, K. Meidani, C. Bruss, M. Goldblum, *Recovering Wasted Compute in Autoresearch Agents* at Conference on Language Modeling (COLM). 2026.

arXiv A. Li, S. McLeish, H. Chen, N. Kalra, **Z. Chen**, et al., *End-to-End Context Compression at Scale*. arXiv preprint, 2026.

ICML 2024 Wksp S. Chang, **Z. Chen**, B. Dumitrascu, D. A. Knowles, *CellFlows: Inferring Splicing Kinetics from Latent and Mechanistic Cellular Dynamics*. ML4LMS Workshop, ICML 2024.

EMNLP 2022 **Z. Chen**, D. Verdi do Amarante, J. Donaldson, Y. Jo, J. Park, *Argument Mining for Review Helpfulness Prediction*. EMNLP 2022.

DSAA 2021 **Z. Chen**, J. Park, *Analyzing Cultural Assimilation through the Lens of Yelp Restaurant Reviews*. IEEE DSAA 2021.

Industry Experience

2021–2024 **Machine Learning Software Engineer**, *CarMax*.

- Built personalized recommendation systems and customer segmentation systems for search engine optimization
- Built microservices, Python libraries, and CI/CD pipelines for data sourcing and model orchestration
- Trained, deployed, and served models at scale via AzureML, Apache Spark, and Databricks

Research Experience

2024–Present **PhD Student**, *Advisors: Prof. Micah Goldblum (EE), Prof. Bianca Dumitrascu (Statistics)*, Columbia University.

- Analyzed LLM training dynamics and identified causes behind pretraining loss spikes
- Developed computation-aware methods for high-dimensional and latent-space Bayesian optimization
- Ran LLM pretraining experiments with models up to 7B parameters across 64 GPUs and TPU VMs using FSDP and tensor parallelism; built tooling for reproducible training diagnostics
- Mentored undergraduates and master's students on autoresearch agents toward a COLM 2026 publication.

2022–2024 **Graduate Research Assistant**, *Advisor: Prof. David Knowles*, Columbia University & New York Genome Center.

- Developed deep-kernel-learning-based active learning to efficiently learn splicing outcomes from DNA sequences
- Trained VAEs and latent neural ODEs to infer cellular dynamics from scRNA-seq data
- Trained DNA language models to understand variant effects on chromatin accessibility

2020–2021 **Honors Thesis & Research Assistant**, *Advisor: Prof. Joonsuk Park*, University of Richmond.

- Implemented transfer learning methods for graph-based argument mining to improve helpfulness prediction of Amazon reviews
- Developed probabilistic graphical models and inference algorithms for topic modeling on online review data for cultural analysis

Teaching & Service

Reviewer ICML, NeurIPS (2026); Nature Machine Intelligence (2023)

Teaching Assistant Statistical Machine Learning (Spring 2025); Advanced Machine Learning (Fall 2024, 2025); Computational Statistics (Spring 2026), Department of Statistics - Columbia University

Technical Skills

ML/AI PyTorch, JAX, distributed / multi-GPU & TPU training, FSDP, TP, SLURM, Weights & Biases

Infrastructure Apache Spark, Databricks, Google Cloud Platform, Microsoft Azure (ML, DevOps, Cosmos DB)

Honors

Phi Beta Kappa; University of Richmond Science Scholar Award (full tuition); National Merit Scholar